

uc3m-thesis-ieee

Package Documentation — v0.4.0

Universidad Carlos III de Madrid thesis template, IEEE style

A Typst template for bachelor/master theses at Universidad Carlos III de Madrid, following university guidelines.

The template is based on `ldcas-uc3m/thesis-template` and `clean-uc3m` (a fork of `clean-dhbw`).

1. Installation

The recommended way to install it is through Typst Universe. You can initialize a new project from the template with:

```
1 typst init @preview/uc3m-thesis-ieee my-final-thesis
```

shell

This creates a `my-final-thesis/` directory with all the files needed to get started.

2. Configuration

The template exports a single `conf` function, which you apply to your document with `#show: conf.with(...)`.

2.1. Minimal example

```
1 #import "@preview/uc3m-thesis-ieee:0.4.0": conf
2
3 #show: conf.with(
4   title: "My Bachelor Thesis",
5   author: "GUL UC3M",
6   degree: "Computer Science and Engineering",
7   advisors: ("Prof. L",),
8   location: "Leganés, Madrid",
9   thesis-type: "TFG",
10  date: datetime(year: 2025, month: 6, day: 15),
11  language: "en",
12  abstract: (
13    body: [A short description of my thesis.],
14    keywords: ("Keyword 1", "Keyword 2", "Keyword 3"),
15  ),
16  genai-declaration: (usage: false),
17 )
18
19 = Introduction
20
21 My thesis starts here.
```

typ

2.2. Abstract

The abstract parameter takes a dictionary with the following keys:

Key	Type	Description
body*	content	The abstract text.
keywords*	array of str	Between 2 and 5 keywords. See IEEE Taxonomy.

```

1  abstract: (
2    body: [
3      This thesis presents an implementation of interrupts and timers
4      in the CREATOR simulator.
5    ],
6    keywords: ("Operating Systems", "Interrupts", "RISC-V", "Simulation"),
7  ),

```

typ

For Spanish documents, it is required to provide an english-abstract with the same structure:

```

1  language: "es",
2  abstract: (
3    body: [Una breve descripción de mi tesis.],
4    keywords: ("Sistemas Operativos", "Interrupciones", "RISC-V"),
5  ),
6  english-abstract: (
7    body: [A brief description of my thesis.],
8    keywords: ("Operating Systems", "Interrupts", "RISC-V"),
9  ),

```

typ

2.3. Epigraph

An optional opening quote placed before the abstract. Provide a dictionary with:

Key	Type	Description
quote*	content	The quote text.
author*	str	The author of the quote.
source	str	(Optional) The source where the quote was found.

```

1  epigraph: (
2    quote: [The purpose of abstracting is not to be vague, but to create a new
3    semantic level in which one can be absolutely precise.],
4    author: "Edsger W. Dijkstra",
5    source: "The Humble Programmer, 1972",
6  ),

```

typ

2.4. Outlines

The outlines parameter accepts an optional dictionary with the following boolean keys. All keys are optional (default false):

Key	Type	Description
figures	bool	Include a list of figures.
tables	bool	Include a list of tables.
listings	bool	Include a list of code listings.
custom	array of content	Additional custom outlines (e.g. algorithms).

```

1  outlines: (
2    figures: true,
3    tables: true,
4    listings: false,
5    // custom: (
6      //  outline(
7        //    title: [List of Algorithms],
8        //    target: figure.where(kind: "algorithm"),
9        //  ),
10   // ),
11 ),

```

2.5. Abbreviations

You can pass abbreviations as a dictionary mapping abbreviations to their expansions (automatically sorted alphabetically):

```

1  abbreviations: (
2    API: "Application Programming Interface",
3    CPU: "Central Processing Unit",
4    ML: "Machine Learning",
5    TFG: "Trabajo de Fin de Grado",
6  ),

```

Or as custom content if you need full control over the layout:

```

1  abbreviations: [
2    #table(
3      columns: (auto, lfr),
4      [*API*], [Application Programming Interface],
5      [*ML*], [Machine Learning],
6    )
7  ],

```

2.6. Glossary

The template integrates with `glossarium` for automatic glossary and acronym management.

Define your entries in a separate file (e.g. `config/glossary.typ`):

```

1  #let glossary-entries = (
2    (
3      key: "API",
4      short: "API",

```

```

5     long: "Application Programming Interface",
6     description: "A set of protocols and tools for building software.",
7   ),
8   (
9     key: "ML",
10    short: "ML",
11    long: "Machine Learning",
12    description: "A subset of AI that enables computers to learn from
13      data.",
14  ),
15 )

```

Then import it in your main file and pass it to conf:

```

1  #import "config/glossary.typ": glossary-entries typ
2
3  #show: conf.with(
4    // ...
5    glossary: glossary-entries,
6  )

```

To reference a glossary term in the body text, use `#gls("key")` (singular) or `#glspl("key")` (plural). These are re-exported from `glossarium` and available after the `#show: conf.with(...)` call:

```

1  #import "@preview/uc3m-thesis-ieee:0.4.0": conf typ
2  #import "@preview/glossarium:0.5.9": gls, glspl
3
4  // In the body:
5  The #gls("API") exposes several endpoints.
6  Multiple #glspl("ML") models were evaluated.

```

Alternatively, you can pass fully custom content to `glossary` if you don't want to use `glossarium`.

2.7. Generative AI Declaration

The `genai-declaration` parameter is **required** by the university. You can either:

2.7.1. Option A: Use the built-in template (recommended)

If you did not use generative AI (`usage: false`):

```

1  genai-declaration: (usage: false), typ

```

If you did use generative AI (`usage: true`), it is required to provide the full declaration:

Pass a dictionary with the following keys:

Key	Type	Required	Description
<code>usage*</code>	<code>bool</code>	Always	Whether you used generative AI in the thesis.

Key	Type	Required	Description
data-usage	dictionary	If usage is true	Data usage declaration. See below.
technical-usage	dictionary	If usage is true	Technical usage declaration. See below.
usage-reflection	content	If usage is true	Personal reflection on AI usage.

data-usage keys — all are bool (true = YES, false = NO):

Key	Description
confidential	Did you submit confidential data (trade secrets, proprietary info)?
copyright	Did you submit copyrighted materials?
personal	Did you submit personal data?
tos	Did your usage respect the tool's terms of use?

technical-usage keys — include only those that apply, with a brief description of the prompt and interaction. Items are automatically grouped into two sections matching the official form:

Documentation and drafting: reflection, review, research, references, summary, translation

Develop specific content: assistance-coding, generating-content, optimization, data-processing, idea-inspiration, other

An optional tool key (str) declares the AI system name and version (e.g. "ChatGPT 4o").

```

1  genai-declaration: (
2      usage: true,
3      data-usage: (
4          // true = YES / used with authorization; false = NO / not used
5          confidential: false,
6          copyright: false,
7          personal: false,
8          tos: true,
9      ),
10     technical-usage: (
11         // Name of the AI system and version (optional):
12         // tool: "ChatGPT 4o",
13
14         // — Documentation and drafting —————
15         reflection: [I asked for a list of alternatives to address the
16             problem stated at the beginning of the paper.],
17         // review: [...],
18         // research: [...],
19         // references: [...],
20         // summary: [...],
21         // translation: [...],
22
23         // — Develop specific content —————

```

```

24     assistance-coding: [I asked for explanations about a Python code
25         snippet and for help correcting an error.],
26     // generating-content: [...],
27     // optimization: [...],
28     // data-processing: [...],
29     // idea-inspiration: [...],
30     // other: [...],
31 ),
32 usage-reflection: [
33     Overall, I found generative AI useful for boilerplate tasks and
34     quick explanations, but it required careful verification of all
35     generated content. It did not replace critical thinking or
36     domain-specific knowledge.
37 ],
38 ),

```

2.7.2. Option B: Provide custom content

If you need full control, pass any content directly:

```

1 genai-declaration: [
2     // Your custom AI declaration content here.
3 ],

```

typ

2.8. Bibliography

Pass the result of Typst's built-in `bibliography()` call:

```

1 // config/bibliography.typ
2 #let bibliography-file = "../references.bib"
3 #let bibliography-style = "ieee"
4 #let bibliography = bibliography(bibliography-file, style: bibliography-
5     style)

```

typ

```

1 // report.typ
2 #import "config/bibliography.typ": bibliography
3
4 #show: conf.with(
5     // ...
6     bibliography-content: bibliography,
7 )

```

typ

3. Styles

The template offers three visual styles, controlled by the `style` parameter.

3.1. "fancy" (default)

An opinionated style inspired by the original LaTeX template, with:

- Decorative chapter title pages with a horizontal rule and chapter number/name centered
- Serif headers showing the current chapter name and page number

- IEEE-style figure captions and table captions
- **Libertinus Serif** body font



Grado en Ingeniería Tipstática

Trabajo de Fin de Grado

«Análisis, diseño, e implementación
del mejor Trabajo de Fin de Grado
de la historia»

Autor

Nombre Extremadamente Largo e Incómodo
de Escribir

Tutor

Profesor Cuyos Padres Tenían Visión de Futuro

Leganés, Madrid
Abril 2025



Esta obra se encuentra sujeta a la licencia Creative Commons
Reconocimiento - No Comercial - Sin Derivadas 4.0 Internacional

CAPÍTULO 1

INTRODUCTION

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Figure 1: Example of the **fancy** style.

3.2. "clean"

A clean, minimal style inspired by clean-dhbw, with:

- Large chapter number in the background on chapter pages
- Simple running header showing the chapter name
- Clean-style figure and table captions
- **Libertinus Serif** body font

uc3m | Universidad
Carlos III
de Madrid

Grado en Ingeniería Tipstática
Trabajo de Fin de Grado

«Análisis, diseño, e
implementación del mejor
Trabajo de Fin de Grado de la
historia»

Autor
Nombre Extremadamente Largo e Incómodo de
Escribir

Tutor
Profesor Cuyos Padres Tenían Visión de Futuro

Leganés, Madrid
Abril 2025



Esta obra se encuentra sujeta a la licencia Creative Commons
Reconocimiento - No Comercial - Sin Derivadas 4.0 Internacional



Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Figure 2: Example of the **clean** style.

3.3. "strict"

Strictly follows the UC3M library guidelines:

- Times New Roman font, 12pt
- Centered uppercase chapter headings
- Page numbers in the footer (Roman numerals in front matter, Arabic in body)
- No decorative elements
- Double-sided layout is **not** permitted with this style



1. INTRODUCTION

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Grado en Ingeniería Telemática

Trabajo de Fin de Grado

«Análisis, diseño, e implementación del
mejor Trabajo de Fin de Grado de la
historia»

Autor

Nombre Extremadamente Largo e Incómodo de
Escribir

Tutor

Profesor Cuyos Padres Tenían Visión de Futuro

Leganés, Madrid
Abril 2025



Este obra se encuentra sujeta a la licencia Creative Commons
Reconocimiento - No Comercial - Sin Derivadas 4.0 Internacional

1

Figure 3: Example of the **strict** style.



Idea

You can mix styles for the title page and body independently using `titlepage-style`. For example, to use a fancy title page with a strict body: `style: "strict", titlepage-style: "fancy"`.

4. API Reference

conf

Main configuration function.

Recommended to use with `#show`: `conf.with(...)`.

Parameters

```

conf(
  doc: content,
  title: str,
  author: str,
  degree: str,
  advisors: array,
  location: str,
  thesis-type: str,
  date: datetime,
  bibliography-content: content,
  language: str,
  style: str,
  titlepage-style: str auto,
  table-style: str auto,
  figure-style: str auto,
  figure-spacing: length none,
  double-sided: bool,
  logo: str,
  short-title: str,
  date-format: str auto,
  license: bool,
  flyleaf: bool,
  epigraph: dictionary none,
  abstract: dictionary,
  english-abstract: dictionary,
  acknowledgements: content none,
  outlines: dictionary none,
  appendixes: content none,
  glossary: array content none,
  abbreviations: dictionary content none,
  genai-declaration: dictionary content
) -> content

```

doc content

Thesis contents.

title str

Title of the thesis.

Default: none

author str

Author name.

Default: none

degree str

Degree name (e.g. "Computer Science and Engineering").

Default: none

advisors array

List of advisor names.

Default: none

location str

Presentation location.

Default: none

thesis-type str

Type of thesis ("TFG" or "TFM").

Default: none

date datetime

Presentation date.

Default: none

bibliography-content content

Bibliography contents, usually calling bibliography.

Default: none

language str

"en" or "es".

Default: none

style `str`

Visual style, mainly affecting headings, headers, and footers. The available styles are `strict`, which strictly follow's the university library's guidelines, `clean`, based on clean-dhbw, and `fancy`, based on my original LaTeX version.

Default: `"fancy"`

titlepage-style `str` or `auto`

Style for the titlepage (see `style`). If set to `auto`, uses the main style.

Default: `auto`

table-style `str` or `auto`

Style for the table caption, either `"clean"` or `"ieee"`. If set to `auto`, uses the default for the main style (`"clean"` for clean style, `"ieee"` for the rest).

Default: `auto`

figure-style `str` or `auto`

Style for the figure caption, either `"clean"` or `"ieee"`. If set to `auto`, uses the default for the main style (`"clean"` for clean style, `"ieee"` for the rest).

Default: `auto`

figure-spacing `length` or `none`

Extra spacing to give to figures and tables. If `none`, no extra spacing.

Default: `0.75em`

double-sided `bool`

Whether to use double-sided pages. This is not allowed in the `strict` style.

Default: `false`

logo `str`

Type of logo (`"old"` or `"new"`).

Default: `"new"`

short-title `str`

Shorter version of the title, to be displayed in the headers. Only applies if `double-sided` is set to `true`.

Default: `none`

date-format `str` or `auto`

Date format. Use `auto` or specify the format using the [Typst format syntax](https://typst.app/docs/reference/foundations/datetime/#format).

Default: `auto`

license `bool`

Whether to include the CC BY-NC-ND 4.0 license.

Default: `true`

flyleaf `bool`

Whether to include a blank page after the cover.

Default: `true`

epigraph `dictionary` or `none`

A short quote that guided you through the writing of the thesis, your degree, or your life. Consists of quote (of type `content`), the body or text itself, author (of type `str`), the author of the quote and, optionally, source (of type `str`), where the quote was found.

Default: `none`

abstract `dictionary`

A short and precise representation of the thesis content. Consists of body (of type `content`), the main text, and keywords, an array of key terms (of type `str`) (see [IEEE Taxonomy](https://www.ieee.org/content/dam/ieee-org/ieee/web/org/pubs/ieee-taxonomy.pdf)).

Default: `none`

english-abstract `dictionary`

An english translation of the abstract. Compulsory for spanish works, invalid for english ones.

Default: `none`

acknowledgements `content` or `none`

Text where you give thanks to everyone that helped you.

Default: `none`

outlines `dictionary` or `none`

Set of extra outlines to include (figures, tables, listings), and extra custom outlines (custom, an array of contents – typically the result of calling `outline`).

Default: `none`

appendixes `content` or `none`

Set of appendixes.

Default: `none`

glossary `array` or `content` or `none`

Glossary entries. If `array` is provided, it will use the `glossarium` library. If `content` is passed, it will display that content, without applying any styling.

Default: `none`

abbreviations `dictionary` or `content` or `none`

Abbreviations, acronyms and initials used throughout the thesis. You can provide a map (dictionary of strings) or a custom one (`content`).

Default: `none`

genai-declaration `dictionary` or `content`

Information about the use of Generative AI in the thesis. You can supply your own content, or use the university's template, by supplying a dictionary. See the example for more details.

Default: `none`